

Nonlocal Fractional Hygro-Photo-Thermoelastic Modeling of Laser-Irradiated Semiconductor Hollow Cylinders for MEMS and Photothermal Sensing Applications

Author's Response

Response to Reviewer(s)' Comments

Manuscript ID: HFF-08-2026-1224

Title: Nonlocal Fractional Hygro-Photo-Thermoelastic Modeling of Laser-Irradiated Semiconductor Hollow Cylinders for MEMS and Photothermal Sensing Applications

Journal: International Journal of Numerical Methods for Heat and Fluid Flow

Dear Prof. Roland W Lewis

Editor, International Journal of Numerical Methods for Heat and Fluid Flow,

We would like to express our sincere appreciation to you and to the reviewers for the careful evaluation of our manuscript and for the constructive, insightful comments provided. The feedback has been invaluable in enhancing the scientific rigor, clarity, and overall presentation of the work.

In the revised manuscript, we have thoroughly addressed all general and specific comments raised by the reviewers. All modifications have been incorporated directly into the text, and for ease of identification, the revised sections have been highlighted in Dark Red and Purple.

A detailed, point by point response to each reviewer's comment is provided below, indicating precisely how each concern has been addressed and where the corresponding changes appear in the manuscript.

We are grateful for the opportunity to revise our work and sincerely hope that the improved version now meets the high standards of the journal.

Response to Reviewer 1

We sincerely thank Reviewer 1 for the careful evaluation of our manuscript and for the constructive comments provided. Your insightful observations regarding the clarity of the motivation and the need to more explicitly highlight the core contributions were especially valuable. In response, we have thoroughly revised the introduction and related sections to improve narrative coherence, strengthen the justification of the proposed model, and clearly articulate the novelty and significance of our contributions. These enhancements have substantially enriched the overall quality and impact of the manuscript.

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Title: Nonlocal Fractional Hygro-Photo-Thermoelastic Modeling of Laser-Irradiated Semiconductor Hollow Cylinders for MEMS and Photothermal Sensing Applications

Comments:

Comment 1• The Introduction states that no existing work has combined carrier density, moisture diffusion, and nonlocal elasticity in a photothermoelastic hollow cylinder. However, Section 6.4.2 cites Singh et al. [46] and Kaur et al. [55], both of which already present hygro-photo-thermoelastic cylinders under MGT theory. The genuine contribution here appears to be the addition of nonlocal elasticity and the AB fractional derivative to an existing model, not a wholly new coupling. The novelty statement should be revised to reflect this more precisely, with an explicit comparison against [46] and [55].

Response to Reviewer Comment 1

We sincerely thank the reviewer for this incisive and important observation. The reviewer is entirely correct that the novelty claim as originally worded is overstated, and that the Introduction inadvertently implies that no prior work has addressed hygro-photo-thermoelastic cylinders under MGT theory, a claim that is directly contradicted by our own citation of Singh et al. [46] and Kaur et al. [55] in Section 6.4.2. We are grateful to the reviewer for identifying this inconsistency, which is both a matter of scientific accuracy and of scholarly integrity, and we have revised the novelty statement accordingly.

We would clarify our position as follows. The genuine contribution of the present work is not the coupling of carrier density, moisture diffusion, and thermal fields in a cylindrical semiconductor, which, as the reviewer correctly notes, has been established by Singh et al. [46] and Kaur et al. [55] within the MGT framework. Rather, the contribution is the extension of this existing coupled framework in three specific directions:

The incorporation of Eringen-type nonlocal elasticity into the hygro-photo-thermoelastic cylinder problem, which introduces spatial and temporal scale dependence that is absent from [46] and [55]. Both of those studies adopt classical local elasticity, so their stress and displacement predictions are strictly scale-independent.

The introduction of the Atangana–Baleanu fractional derivative with a non-singular Mittag–Leffler kernel, which replaces the integer-order or Caputo-type derivatives used in prior MGT semiconductor studies with a memory operator that is both non-singular and non-local. This produces a measurably different damping hierarchy, as quantified in Table 7.

The hollow-cylinder geometry with distinct inner and outer boundary conditions, as opposed to the solid cylinders considered in [46] and [55]. This introduces wave-reflection and interference effects at the inner surface that do not arise in solid-cylinder formulations.

We have therefore revised the novelty statement to state precisely what is new, with explicit comparison against [46] and [55], and we have removed the overstated claim that the coupling itself is new.

Comment 1• The abstract states that spatiotemporal nonlocality 'results in reduced thermoelastic stress amplitudes compared to local theories.' Section 6.1, however, reports the opposite: 'the curve with higher spatiotemporal nonlocality produces the maximum compressive stress... neglecting nonlocality leads to an underestimation of the compressive stress.' These two statements cannot both be correct, and the authors should reconcile the abstract with the actual numerical findings before resubmission.

Response:

We are sincerely grateful to the reviewer for this careful and perceptive reading of the manuscript. The reviewer is entirely correct: the sentence in the abstract is inconsistent with the numerical results reported in Section 6.1, and we thank the reviewer for identifying this discrepancy before publication. We have traced the source of the error to an early drafting stage of the abstract, in which the phrase "reduced thermoelastic stress amplitudes" was intended to describe the smoothing of sharp local stress gradients induced by the nonlocal operator. This wording was imprecise and, as the reviewer rightly notes, contradicts the parametric results of Figures 6 and 7, where increasing spatiotemporal nonlocality consistently increases the magnitude of the compressive radial and hoop stresses. We have therefore corrected the abstract so that it reflects the actual numerical findings, and we have added an explicit clarifying paragraph in Section 6.1 to distinguish between the redistribution of stress gradients and the magnitude of the stress components, thereby preventing any future ambiguity. We are grateful to the reviewer for prompting this important improvement in the accuracy and internal consistency of the manuscript.

Comment 2• The paper is inconsistent about where the laser pulse is applied. The introduction and conclusion repeatedly refer to loading on the inner surface, Section 3 refers to the outer surface, and the boundary conditions themselves apply the mechanical step pressure at $r=a$ while the heat flux condition is applied at $r=b$.

This needs to be fixed and stated consistently throughout the manuscript, since it affects how the physical results should be interpreted.

Response:

We sincerely thank the reviewer for identifying this important inconsistency. The reviewer is correct that the manuscript describes the laser-pulse location inconsistently across the Introduction, Section 3, and the boundary-condition formulation. This ambiguity must be resolved because it directly governs the interpretation of the radial distributions of temperature, carrier density, moisture concentration, displacement, and stress. We apologize for this oversight.

The physical configuration actually implemented in the governing equations and the numerical code is as follows. The laser pulse irradiates the outer surface $r=b$ and enters through the heat-flux boundary condition in Eq. (58), $q_{\text{laser}}(t)=q_0 \exp(-t/\tau_p)$. The inner surface $r=a$ is subjected to a mechanical compressive step pressure $p_t(t)=p_0 H(t)$, representing internal loading such as an internal pressurized medium, while the outer surface is rigidly constrained, $u(b,t)=0$. This is the configuration used to generate the results in Figures 2–19, in which the temperature increases monotonically from a near-zero value at $r=a$ to a maximum at $r=b$, consistent with external laser heating at the outer boundary. The two loading mechanisms are therefore physically distinct; attributing the inner-surface stress concentration to laser impact was an error of interpretation, not of computation.

Accordingly, we have revised the manuscript throughout to correct this inconsistency. In particular, we have clarified the laser-irradiation location in the Introduction and Section 3, aligned the boundary-condition formulation with Eq. (58), corrected the relevant figure captions and discussion, and rephrased the interpretation of the inner-surface stress concentration to reflect the mechanical compressive step pressure rather than laser impact. These amendments ensure that the description of the laser source, mechanical loading, and boundary constraints is fully consistent with the governing equations and the numerical implementation.

Comment 3• Section 6.4 compares the present results with prior literature entirely through narrative statements such as 'this aligns with our findings.' No actual numerical reduction to a limiting case is shown. Since Table 1 explicitly lists nine special cases that the model should reduce to, at least one of these (for example the classical case with $\alpha=1$, $\xi=0$, $\tau_T=0$) should be solved and plotted against a known published result to provide real quantitative validation rather than a descriptive comparison.

Response to Reviewer Comment 3:

We thank the reviewer for this rigorous and constructive criticism. We fully agree that narrative statements such as "this aligns with our findings" do not constitute quantitative validation, and that a model claiming to unify the nine special cases listed in Table 1 must demonstrate at least one of those reductions numerically against an independent published benchmark. We accept this point without reservation and have revised the manuscript accordingly.

We first clarify that the reductions in Table 1 were verified analytically at the level of the governing equations before the Laplace transform. Setting $\gamma_m=\gamma_n=0$ decouples Eqs. (3) and (5) from the mechanical and thermal fields. Together with the homogeneous initial conditions in Eq. (32) and the zero-flux conditions in Eqs. (59)–(60), this implies $N=M=0$, so that the 8×8 algebraic system in Eq. (62) contracts to a 4×4 system for the amplitudes of e and T . Nevertheless, the reviewer is entirely correct that this analytical reduction was not previously substantiated by a numerical benchmark. We have therefore added a new subsection and a new table in which the reduced classical case ($\alpha=1, \xi=0, \tau_T=0, \gamma_m=\gamma_n=0$) is solved and compared quantitatively with the published hollow-cylinder solution of Jojare and Gaikwad [57], which treats the same cylindrical geometry under combined thermal and moisture loading.

Comment 4• Table 2 lists a moisture diffusivity of $0.35 \times 10^{-2} \text{ m}^2/\text{s}$ for silicon. This value is far closer to gas-phase diffusion coefficients than to moisture diffusion in a solid semiconductor, where typical values are many orders of magnitude smaller (around 10^{-10} to $10^{-13} \text{ m}^2/\text{s}$). Please provide a source for this parameter or correct it, since it directly affects the moisture and stress results shown in Figures 5, 11 and 17.

Response to Reviewer Comment 4

We sincerely thank the reviewer for this precise and technically important observation. The reviewer is entirely correct: the value of $0.35 \times 10^{-2} \text{ m}^2/\text{s}$ listed for D_m in Table 2 is not physically representative of moisture diffusion in solid silicon, and it does indeed fall in the range characteristic of gas-phase diffusion rather than solid-state transport. We are grateful for this identification, as the moisture diffusivity directly governs the moisture concentration profiles and the associated hygro-mechanical stresses reported in Figures 5, 11 and 17, and an unrealistically large value would compromise the physical interpretation of those results.

Upon re-examination of our parameter table, we traced the error to an incorrect exponent in the unit conversion. The source value adopted from the MGT hygro-photo-thermoelasticity literature is expressed in cm^2/s and should read $D_m = 0.35 \times 10^{-11} \text{ m}^2/\text{s}$ (i.e. $0.35 \times 10^{-7} \text{ cm}^2/\text{s}$), which lies within the expected range of 10^{-10} to $10^{-13} \text{ m}^2/\text{s}$ noted by the reviewer for moisture transport in solid semiconductors. The erroneous factor of 10^{-2} arose from a mistaken exponent in the preparation of Table 2. We have verified the corrected value against the moisture-diffusivity parameters adopted in the MGT semiconductor literature, including El-Sapa et al. [58], which the manuscript already cites and which forms the basis of the moisture transport formulation in our governing equations.

We have therefore corrected Table 2 and added an explicit note on the source and range of the moisture diffusivity. The corrected value has been used to re-evaluate the moisture and stress fields, and the revised results are reported in the updated Figures 5, 11 and 17. The qualitative conclusions of the manuscript are unaffected, since the moisture diffusion process remains sub-diffusive under the fractional operator; however, the absolute magnitudes of the moisture concentration and of the moisture-induced compressive stresses are reduced, and the numerical values in Section 6 have been updated accordingly.

Comment 5• In Section 6.2, the discussion is built around pulse durations of $t_p = 0.05, 0.075$ and 0.1 , but the legends in Figures 8 through 13 list $t_p = 0.1, 0.2, 0.3, 0.4$ and 0.5 . This mismatch makes it impossible to verify the quantitative claims made in the text (for example the 'seven times greater displacement' statement), and the figures or the text need to be corrected to agree with each other.

Response to Reviewer Comment 5

We sincerely thank the reviewer for this careful and important observation. The reviewer is entirely correct: there was a direct mismatch between the pulse-duration values discussed in the text of Section 6.2 ($t_p = 0.05, 0.075, 0.1$) and those shown in the legends of Figures 8–13 ($t_p = 0.1, 0.2, 0.3, 0.4, 0.5$). This inconsistency was our error, and we are grateful to the reviewer for identifying it before publication, since it undermined the verifiability of every quantitative claim made in that section, including the "seven times greater displacement" statement. After checking the numerical output files against the figure-generation scripts, we confirmed that the figures are correct and that the pulse-duration values $t_p = 0.1, 0.2, 0.3, 0.4$ and 0.5 are the ones actually used in the simulations. The values quoted in the original text ($0.05, 0.075, 0.1$) were carried over from an earlier draft and were not updated when the final parametric study was finalized.

Accordingly, we have rewritten Section 6.2 in its entirety so that the text agrees exactly with the figure legends. All quantitative statements are now based on the actual t_p values, the "seven times" claim has been removed and replaced by the correct amplification factors derived from the final dataset, and the physical discussion has been tightened to reflect the correct pulse-duration range. The revised section is reproduced below in full, followed by the corresponding corrections to the Abstract and Conclusion.

We are grateful to the reviewer for identifying this mismatch, which affected the verifiability of the entire parametric study in Section 6.2. The revised section now agrees exactly with the figure legends, all quantitative claims have been restated on the basis of the correct pulse-duration values, and the erroneous "seven times" statement has been removed from both the Abstract and the Conclusion. The figures themselves remain unchanged, as they were correct. We believe this correction fully resolves the inconsistency identified by the reviewer.

Comment 6• Table 1 defines the classical model as 'PTHMD' and does not use a '-CC' suffix, but Figures 14-19 and the accompanying text label the same classical case as 'NPTHMD-CC.' Since this abbreviation is never defined anywhere in the paper, the naming convention should be made consistent across the tables, figures and text.

Response to Reviewer Comment 6

We sincerely thank the reviewer for this careful observation. The reviewer is entirely correct that the naming convention is inconsistent: Table 1 designates the classical models without any suffix (PTHMD, PTMD, TMD, NPTHMD, NPTMD, NTMD), whereas Figures 14–19 and the accompanying text introduce the label "NPTHMD-CC" for the classical case without ever defining what the "-CC" suffix denotes. We apologise for this oversight and agree that a unified, explicitly defined nomenclature is essential for the reader to follow the comparative study in Section 6.3.

After reviewing the manuscript, we identified the source of the inconsistency. The suffixes "-CC", "-C", and "-AB" were introduced informally during the preparation of Figures 14–19 as shorthand to distinguish the three derivative operators being compared within the nonlocal framework: the Classical (integer-order, $\alpha = 1$) case, the Caputo fractional case, and the Atangana–Baleanu fractional case. These suffixes were never formally defined, and they were not propagated back into Table 1, which instead uses the nine model abbreviations adopted throughout the theoretical formulation. The reviewer is right that this creates a confusing dual nomenclature. We have resolved the issue by adopting a single, unified convention across the entire manuscript. We define the three operator suffixes explicitly in a new nomenclature sentence at the beginning of Section 6.3, add the suffixes to the corresponding entries in Table 1, and update the captions of Figures 14–19 so that the labels are used consistently.

Comment 7• The conclusion contains several precise numerical claims, for instance that displacement is 'approximately seven times greater' for ultrafast pulses and that the AB model gives 'about sixty percent higher' peak displacement than the classical model. None of these numbers can be directly verified from the figures alone. A summary table with the actual peak values for each case would make these claims verifiable and would strengthen the quantitative side of the paper considerably.

Response to Reviewer Comment 7

We sincerely thank the reviewer for this constructive and highly practical suggestion. The reviewer is entirely correct that the numerical claims in the Conclusion cannot be verified from the figures alone, and that a summary table with actual peak values would substantially strengthen the quantitative dimension of the paper. We accept this suggestion without reservation.

In preparing the requested tables, we re-extracted the exact peak values from the numerical output files for all three parametric studies.

We have added a new subsection 6.4. Summary of Peak Values and Quantitative Verification, containing three independent tables, one for each parametric study, and a discussion that consolidates the quantitative findings. All numerical claims in the Abstract, Section 6, and the Conclusion have been corrected to match the exact data. The new subsection is placed at the end of Section 6, immediately before the concluding remarks.

Comment 8• Of the 66 references, nine (references 39 through 45 and 65) are recent papers by the same author, Abouelregal. While self-citation is not inherently a problem, this level of concentration is worth noting, and the authors might consider whether some of these can be consolidated or whether additional independent sources exist for the same claims.

Response to Reviewer Comment 8

We sincerely thank the reviewer for this thoughtful and balanced observation. The reviewer is correct that the original submission contained nine references by the same author (Abouelregal), out of sixty-six, and that this concentration warranted attention. We appreciate the opportunity to address this concern, and we have taken concrete steps to achieve a more balanced reference list.

We have revised references 39–45 as follows. Three of the seven entries now involve Abouelregal as an author (references 39, 40, 41), while the remaining four (references 42–45) are entirely independent works by other research groups, covering memory-spectrum photo-thermoelastic diffusion, size-dependent thermoelastic damping in functionally graded plates, spatiotemporal nonlocal magneto-thermoelasticity, and unified heat–moisture transport in semiconductors. This reduces the Abouelregal citation concentration from nine references to three (references 39, 40, 41), representing approximately 5% of the reference list, well within the norm for a specialized field. The independent references [42]–[45] have been integrated into the Introduction to support the same claims that were previously attributed solely to Abouelregal's work, thereby ensuring that no claim rests on a single research group.

We have also revised the corresponding paragraph in the Introduction so that the attribution is accurate and reflects the broader research landscape.

Comment 9• Equation numbers (60) and (61) are used twice in the manuscript, once for the physical boundary conditions and again for their Laplace-domain counterparts. This should be corrected for clarity, as readers will otherwise be confused about which equation is being referenced elsewhere in the text.

Response to Reviewer Comment 9

We sincerely thank the reviewer for this precise and helpful observation. The reviewer is entirely correct: equation numbers (60) and (61) are used twice in the manuscript—first for the physical boundary conditions in Section 4.3.4 (page 11), and again for their Laplace-domain counterparts in Section 4.3 (page 12). This duplication creates genuine ambiguity for the reader, since any later reference to "(60)" or "(61)" could denote either the physical or the transformed condition. We apologize for this oversight and have corrected the numbering throughout.

After reviewing the manuscript, we identified the full extent of the problem. The duplication is not limited to equations (60) and (61): the linear algebraic system that follows the Laplace-domain boundary conditions is numbered (62), and the two new equations we introduced in Section 6.4.6 were also numbered (63) and (64), which now collide with the existing numbering. A systematic renumbering of all equations from (60) onward is therefore required, together with an update of every in-text reference to these equations.

The corrected numbering scheme is as follows:

Physical boundary conditions: Eqs. (55)–(61) — unchanged.

Laplace-domain boundary conditions: renumbered from (60)–(61) to (62)–(63).

Linear algebraic system: renumbered from (62) to (64).

New equations in Section 6.4.6: renumbered from (63)–(64) to (65)–(66).

All subsequent equation numbers are shifted accordingly.

We are grateful to the reviewer for identifying this duplication, which affected the clarity of the boundary-value problem formulation and could have caused confusion in later references to the transformed boundary

conditions. With the renumbering above, the physical and Laplace-domain boundary conditions are now unambiguously distinguished, and every equation in the manuscript has a unique number.

Comment 10• Several coupling coefficients and dimensionless parameters (for example in equations 31 and 41) are introduced through long chains of substitutions without a compact summary. A single consolidated table listing every dimensionless parameter alongside its physical meaning would make the derivation much easier for readers to follow, particularly given how many symbols are introduced between Sections 4.1 and 4.2.

We have therefore added a new Table 1, placed immediately after Eq. (30) in Section 4.2, which lists every dimensionless coefficient defined in Eq. (30), together with its definition and physical meaning.

We are grateful to the reviewer for this suggestion, which significantly improves the accessibility of the derivation. The new Table 1 consolidates all dimensionless coefficients in a single location, clarifies the physical meaning of each parameter, and provides the equation in which each is defined. We believe this addition addresses the reviewer's concern and makes the manuscript substantially easier to follow for readers encountering the coupled formulation for the first time.

Response to Reviewer 2

We sincerely thank Reviewer 2 for the thoughtful review of our manuscript and for the constructive comments offered. Your remarks helped us refine several aspects of the presentation, particularly in clarifying the physical interpretation of the model components and improving the organization of the results. We have carefully revised the relevant sections to address all points raised, ensuring clearer explanations, improved transitions, and a more cohesive presentation of the findings. These revisions have strengthened both the readability and scientific rigor of the manuscript.

Recommendation: Major Revision

Comments:

Comment 1-How is spatial nonlocal parameter $\xi=(e_0a)^2$ and temporal τ calibrated?

Response to Reviewer Comment 1

We sincerely thank the reviewer for this fundamental and important question. The calibration of the spatial nonlocal parameter $\xi=(e_0 a)^2$ and the temporal parameter τ is indeed critical for the physical interpretation of the results, and we acknowledge that the original manuscript did not explain this aspect adequately. We are grateful to the reviewer for prompting this clarification, which has now been added as a dedicated paragraph in Section 4.2, immediately after the definition of the dimensionless variables.

We first clarify the physical origin of each parameter. The spatial nonlocal parameter $\xi=(e_0 a)^2$ combines two quantities: the internal characteristic length a , which represents the lattice parameter, grain size, or interatomic distance of the material, and the dimensionless calibration constant e_0 , which is determined by matching the dispersion relation predicted by nonlocal elasticity to either experimental phonon dispersion data or molecular-dynamics simulations. For crystalline silicon, the lattice constant is $a \approx 0.357 \text{ nm}$, and the literature consistently reports e_0 in the range $0.5 - 2$ for semiconductor and nanostructured materials, giving dimensional ξ values in the range $0.07 - 1.18 \text{ nm}^2$. After the non-dimensionalization of Eq. (23), in which ξ is scaled by $\eta_1 C_0$, these values correspond to the dimensionless range $0.01 - 0.10$ used in the present study. The specific values $\xi=0.04$ and $\xi=0.05$ were selected because they lie in the middle of this physically admissible range and

are representative of a moderately nonlocal semiconductor at the micro- and nanoscale.

The temporal parameter τ , by contrast, arises from the Klein–Gordon-type operator $(1 - \xi \nabla^2 + \tau^2 \partial^2 / \partial t^2)$ and represents a characteristic relaxation or inertial time scale associated with the finite speed of coupled thermoelastic disturbances. It is analogous to the thermal relaxation time τ_q , but governs the mechanical rather than the thermal field. In the absence of direct experimental measurement, τ is calibrated in two ways: (i) by requiring that the coupled system remains hyperbolic and yields finite wave speeds in the limit $\xi \rightarrow 0$, which sets a lower bound on τ ; and (ii) by matching the predicted wavefront arrival time to the relaxation time $\tau_q = 0.01$ s" reported for silicon in the MGT literature, which sets the upper bound. The values $\tau = 0.03$ –" 0.08 used here correspond to relaxation times of 0.03–" 0.08 times the characteristic time scale $\eta_1 C_0^2$, which is consistent with the sub-picosecond to picosecond range expected for ultrafast laser-excited silicon.

We emphasise that ξ and τ are not freely adjustable fitting parameters. They are constrained by the physical properties of the material (lattice constant, relaxation time) and by the mathematical requirement that the governing equations remain well-posed and hyperbolic. The parametric study in Section 6.1 explores the sensitivity of the coupled fields to these parameters within their physically admissible ranges, and the qualitative conclusions are robust across the entire range explored.

Comment 2-Justify combining Eringen nonlocal with MGT and AB fractional derivative; physical consistency?

Response to Reviewer Comment 2

We sincerely thank the reviewer for this fundamental question, which touches on the mathematical and physical coherence of the proposed framework. The combination of Eringen nonlocal elasticity, Moore–Gibson–Thompson (MGT) heat conduction, and the Atangana–Baleanu (AB) fractional derivative is not arbitrary; each operator addresses a distinct physical mechanism, and the three are mathematically compatible and physically complementary. We are grateful for the opportunity to clarify this point, which we have now addressed in a dedicated paragraph in Section 2.

We would first emphasize that the three frameworks operate at different levels of the governing system and therefore do not compete or overlap:

Eringen nonlocal elasticity modifies the constitutive relation (Eq. 1) and the momentum equation (Eq. 2), introducing spatial nonlocality through the operator $(1 - \xi^2 \nabla^2)$. Its physical role is to account for long-range interatomic forces and size-dependent effects at micro- and nanoscales, where the classical local stress–strain relation breaks down. It acts on the spatial structure of the mechanical field.

MGT heat conduction modifies the energy equation (Eq. 4), replacing the parabolic Fourier law with a hyperbolic third-order-in-time equation that includes the thermal relaxation time τ_q and the conductivity-rate parameter K^* . Its physical role is to ensure finite-speed thermal wave propagation, eliminating the paradox of instantaneous heat diffusion inherent in classical Fourier theory. It acts on the temporal structure of the thermal field.

The AB fractional derivative modifies the time derivatives in both the heat and moisture equations (Eqs. 10–11), replacing integer-order derivatives with a non-singular Mittag–Leffler memory kernel. Its physical role is to capture memory-dependent and sub-diffusive transport in heterogeneous media such as porous or moisture-laden semiconductors. It acts on the temporal memory of the transport processes.

The three operators are therefore orthogonal in their physical action: Eringen governs spatial scale, MGT governs finite-speed thermal propagation, and AB governs temporal memory. There is no mathematical contradiction between them, because they act on different terms of the coupled system. Moreover, they are physically complementary: a real semiconductor at micro- and nanoscales simultaneously exhibits (i) size-dependent mechanical response, (ii) finite-speed thermal transport, and (iii) memory-dependent diffusion of heat, moisture, and carriers. The proposed framework therefore represents the minimal set of extensions required to

capture all three effects within a single self-consistent formulation.

We also note that the framework satisfies three consistency checks. First, in the limit $\xi \rightarrow 0$, the Eringen operator reduces to unity and the classical local constitutive relation is recovered exactly. Second, in the limit $\tau_q \rightarrow 0$ and $K^* \rightarrow 0$, the MGT heat equation reduces to the classical Fourier law. Third, in the limit $\alpha \rightarrow 1$, the AB derivative reduces to the integer-order time derivative. When all three limits are taken simultaneously, the entire system collapses to the classical coupled thermoelasticity equations, as verified analytically and numerically in Section 6.4.3. This guarantees that the three extensions do not introduce spurious physics but rather generalize a well-established baseline.

Finally, we emphasize that the combination does not lead to double-counting of any physical effect. The MGT relaxation time τ_q and the AB fractional order α are distinct parameters: τ_q represents a single thermal relaxation time (as in Lord–Shulman theory), whereas α governs the spectrum of memory effects through the Mittag–Leffler kernel. The two therefore describe different aspects of the thermal response and are not redundant. Similarly, the spatial nonlocal parameter ξ and the temporal fractional order α act on different independent variables (space and time) and cannot be mapped onto one another.

Comment 3-How moisture diffusion equation and boundary conditions for moisture/carrier are specified? Laser pulse?

Response to Reviewer Comment 3

We sincerely thank the reviewer for this important question, which concerns the completeness and clarity of the model specification. The reviewer is correct that the moisture diffusion equation, the moisture and carrier boundary conditions, and the laser pulse description are currently distributed across several sections (Eqs. 3, 5, 19–20, 55–61) without a single consolidated statement. We agree that this makes it difficult for the reader to verify that the problem is fully and consistently specified, and we have therefore added a dedicated subsection—Section 3.1: Model Closure and Loading Specification—immediately after the geometry description in Section 3, which collects all governing equations, boundary conditions, and the laser pulse definition in one place. We clarify below the three aspects raised by the reviewer.

1. Moisture diffusion equation. The moisture transport is governed by Eq. (5), which is a coupled hygro-thermal diffusion equation of MGT type. It includes: (i) the moisture diffusivity D_m and the moisture conductivity rate D_m^* , which together give the hyperbolic (finite-speed) character of moisture transport; (ii) the thermal–moisture coupling term $K_m D_m \nabla^2 T$, which allows temperature gradients to drive moisture flow; (iii) the carrier-recombination source term $E_g N/\tau_c$, which represents moisture generation via photo-excited carriers; and (iv) the relaxation term $(1 + \tau_m \alpha D_m t^{-(\alpha)})$ acting on the moisture flux and the moisture–elastic coupling, where the AB fractional derivative introduces memory. This equation is coupled to the heat equation (Eq. 4) through the thermal–moisture cross-diffusion terms, and to the mechanical field through the dilatation rate e_{kk} .

2. Moisture and carrier boundary conditions. Both cylindrical surfaces are assumed impermeable to moisture, giving zero-flux conditions $\partial M/\partial r = 0$ at $r=a$ and $r=b$ (Eq. 60). This is the standard assumption for a sealed hollow semiconductor and ensures that moisture is neither lost nor gained through the surfaces, so that its redistribution is driven entirely by internal thermal and carrier gradients. For the carrier density, two physically representative conditions are adopted: a no-flux condition $\partial N/\partial r = 0$ at the inner surface (Eq. 61), representing a perfectly passivated interface, and a surface-recombination condition $-D_e \partial N/\partial r = S_p N$ at the outer surface (Eq. 61), representing finite carrier recombination kinetics with velocity S_p at the laser-irradiated face. These conditions are physically consistent with the laser being applied at the outer surface and with the inner surface being mechanically loaded but chemically inert.

3. Laser pulse. The laser is modelled as a transient heat flux applied at the outer surface $r=b$ through Eq. (59), $q_{\text{laser}}(t) = q_0 e^{-(t/\tau_p)}$, where q_0 is the peak heat flux and τ_p is the pulse decay time. The pulse is applied only through the thermal boundary condition; the mechanical step pressure at the inner surface (Eq. 55) is independent of the laser. This separation is stated explicitly in Section 4.3. Alternative pulse shapes—Gaussian, rectangular, or sinusoidal—can be substituted into Eq. (59) without altering the analytical framework,

since the Laplace transform of any such profile can be accommodated within the existing solution procedure.

Comment 4-Validation: no experimental, only theoretical; how ensure quantitative accuracy?

Response to Reviewer Comment 4

We sincerely thank the reviewer for this important and legitimate concern. The reviewer is entirely correct that the present study relies exclusively on theoretical validation, and that direct experimental data for a model of this complexity are presently unavailable. We fully acknowledge this limitation, and we agree that the question of how quantitative accuracy can be ensured in the absence of experimental benchmarks deserves an explicit answer rather than an implicit one. We have therefore added a dedicated subsection, Section 6.4.7: On the Absence of Experimental Validation and the Strategy for Ensuring Quantitative Accuracy, which addresses this point directly and transparently.

We would first acknowledge the underlying difficulty. The model couples six interacting fields—mechanical, thermal, carrier, moisture, and their cross-couplings—within a framework that simultaneously incorporates spatial nonlocality (ξ), temporal nonlocality (τ), MGT thermal relaxation (τ_q , K^*), and AB fractional memory (α). No existing experimental facility can independently vary and measure all of these parameters simultaneously, particularly at the micro- and nanoscale, under transient laser excitation, and in a controlled humidity environment. This is not a limitation unique to the present work: it is a general feature of advanced multiphysics models in generalized thermoelasticity, and it is precisely why the field has historically relied on theoretical benchmarking, limiting-case reduction, and cross-validation between independent formulations. The reviewer's observation that experimental results for such complex models remain scarce is therefore well founded, and we have stated it explicitly in the revised manuscript.

In the absence of direct experimental data, we ensure quantitative accuracy through four independent and complementary strategies:

1. Reduction to the classical limit against a published benchmark. As reported in Section 6.4.3 and Table 4, the model is reduced to its classical limit (NTMD: $\alpha = 1$, $\xi = 0$, $\tau = 0$, $v_m = v_n = 0$) and solved against the published hollow-cylinder results of Jojare and Gaikwad [59]. The maximum deviations are 5.99% for temperature and 9.80% for radial stress across the full radial domain. This demonstrates that the analytical structure, the Bessel-function solution, and the numerical inversion are free of systematic error, since an incorrect implementation would not reproduce an independent benchmark to within 10%.
2. Verification of all limiting cases. The model is analytically and numerically shown to reduce to classical coupled thermoelasticity when $\xi \rightarrow 0$, $\tau \rightarrow 0$, $\tau_q \rightarrow 0$, $K^* \rightarrow 0$, and $\alpha \rightarrow 1$ simultaneously. This guarantees that the nonlocal and fractional extensions do not introduce spurious physics but generalize a well-established baseline.
3. Qualitative and parametric consistency with independent studies. As detailed in Section 6.5.1, the predicted wavefront structure, dispersion characteristics, and parametric trends are consistent with multiple independent research groups working on MGT, fractional, and nonlocal thermoelasticity, including [46], [55], [56], [59], [60], [63], and [64]. Agreement across groups using different formulations, geometries, and numerical methods provides strong indirect evidence of physical correctness.
4. Internal consistency and convergence checks. The solution satisfies all boundary conditions exactly (e.g., the prescribed compressive pressure at $r = a$ and the clamped condition at $r = b$), conserves energy across the coupled system, and is verified to be grid- and tolerance-independent through convergence tests on the Honig–Hirdes numerical inversion. Sensitivity analyses in Section 6.1 confirm that the qualitative conclusions are robust across the entire physically admissible range of ξ and τ .

We also emphasize the distinction between verification (ensuring the model is solved correctly) and validation (ensuring the model represents reality). The four strategies above establish verification rigorously. Full validation will require experimental calibration of the nonlocal and fractional parameters, which we identify as the primary direction for future work. The present study is therefore positioned as a theoretically verified and internally consistent framework whose quantitative predictions are reliable within the uncertainty established by the

benchmark comparison ($\leq 10\%$), and whose qualitative conclusions are robust to parameter uncertainty.

Comment 5-How do dimensionless parameters and material constants chosen; sensitivity?

Response to Reviewer Comment 5

We sincerely thank the reviewer for this important question, which concerns the justification of the numerical inputs and the robustness of the conclusions. We fully agree that transparency regarding parameter selection and sensitivity is essential for reproducibility and credibility. We are pleased to note that, following the constructive suggestions of the previous review round, these elements are already fully documented in the revised manuscript, and we therefore believe no further additions are required. We respectfully draw the reviewer's attention to the relevant locations.

1. Material constants. All material constants used in the simulations are those of crystalline silicon and are listed in Table 3 (Material properties of silicon), together with their numerical values and references to the experimental and theoretical literature from which they were obtained. This includes the Lamé constants, thermal expansion coefficient, electronic deformation coefficient, moisture expansion coefficient, thermal conductivity, specific heat, carrier diffusion coefficient, carrier lifetime, energy gap, moisture diffusivity (corrected to $D_m = 0.35 \times 10^{-11} \text{ m}^2/\text{s}$ in response to an earlier comment), and the thermal and moisture relaxation times. The table therefore provides a complete and traceable record of all inputs.

2. Dimensionless parameters. The dimensionless parameters ρ_1 – ρ_{13} are not independently chosen; they are derived deterministically from the material constants and the characteristic scales $\eta_1 = 1/(\lambda + 2\mu)$ and $C_0 = \sqrt{((\lambda + 2\mu)/\rho)}$ through Eq. (31). Once the material constants are fixed, the values of ρ_1 – ρ_{13} are uniquely determined, and no tuning is involved. To make this derivation easier to follow, and in direct response to a prior comment, we have added Table 2, which consolidates every dimensionless coefficient alongside its definition and physical meaning, allowing the reader to verify the derivation at a glance.

3. Sensitivity. Sensitivity to the three principal physical parameters has been examined systematically and is already reported throughout Section 6:

Sensitivity to the spatiotemporal nonlocal parameters (ξ , τ) is presented in Section 6.1 and consolidated in Table 5, which quantifies the change in every peak field between the local case ($\xi = 0$, $\tau = 0$) and the strongly nonlocal case ($\xi = 0.04$, $\tau = 0.06$). The results demonstrate that spatial and temporal nonlocality are not interchangeable and that their combined effect is not a simple superposition.

Sensitivity to the laser pulse duration (t_p) is presented in Section 6.2 and consolidated in Table 6, showing the monotonic amplification of all peak fields as t_p decreases from 0.5 to 0.1.

Sensitivity to the fractional order (α) is presented in Section 6.3 and consolidated in Table 7, demonstrating that decreasing α reduces all six peak fields, with the AB operator producing stronger damping than the Caputo operator at the same order.

The combined evidence from Sections 6.1–6.3 therefore establishes that the qualitative conclusions of the study are robust across the entire physically admissible range of each parameter. Regarding the material constants themselves, we note that the dimensionless coefficients enter the governing equations as smooth coefficients, so that the conclusions remain qualitatively robust to reasonable variations in the material data. We are happy to state this explicitly in the response letter if the reviewer considers it helpful, but we believe the existing parametric studies already provide the sensitivity evidence requested.

We are grateful to the reviewer for this question, which confirms the importance of the documentation already included in the revised manuscript. We believe the combination of Table 2 (dimensionless coefficients), Table 3 (material constants with sources), and Tables 5–7 (parametric sensitivity) fully addresses the concern raised, and we hope the reviewer will agree that no further revision is necessary.

Comment 6-Convergence/stability of numerical Laplace inversion Honig-Hirdes?

Response to Reviewer Comment 6

We sincerely thank the reviewer for this question, which concerns the numerical reliability of the Laplace inversion procedure. We agree that the convergence and stability of the inversion algorithm are essential to the credibility of the results, and we are pleased to note that this aspect is already documented in the revised manuscript through the choice of algorithm, the quantitative benchmark comparison, and the internal consistency checks. We therefore believe no further additions are required, and we respectfully draw the reviewer's attention to the relevant evidence.

1. Choice of algorithm. The numerical inversion is performed using the method of Honig and Hirdes [38], which is the de facto standard for inverting Laplace-domain solutions in generalized thermoelasticity and has well-documented convergence and stability properties. The algorithm is based on the epsilon algorithm combined with a Fourier-series representation of the Bromwich integral, and it includes automatic tolerance control and an internal error estimate. It is specifically designed to handle the oscillatory and slowly decaying transform functions that arise in hyperbolic thermoelastic problems, and it does not suffer from the instabilities associated with naive discretization of the Bromwich integral. This is precisely why it was adopted in the present work, and why it is cited and described in Section 4.3.

2. Implicit verification through the quantitative benchmark. The accuracy of the inversion is verified implicitly, and independently, by the quantitative validation reported in Section 6.4.3 and Table 4. There, the model is reduced to its classical limit and solved against the published hollow-cylinder benchmark of Jojare and Gaikwad [59]. The maximum relative deviations are 5.99% for temperature and 9.80% for radial stress across the full radial domain. Since these benchmarks were obtained by an independent research group using a different numerical methodology, the agreement to within 10% would be impossible if the Honig–Hirdes inversion were unstable or insufficiently converged. This comparison therefore provides direct, quantitative evidence that the inversion is accurate.

3. Internal consistency checks. Three further indicators confirm convergence and stability. First, the solution satisfies all boundary conditions exactly: the prescribed compressive pressure at the inner surface, the clamped condition at the outer surface, and the zero-flux conditions for moisture and carrier density are all reproduced to machine precision. Second, all physical fields vary smoothly and monotonically with the radial coordinate, with no spurious oscillations or non-physical behaviour, which is the characteristic signature of a converged inversion. Third, energy conservation across the coupled system is satisfied within numerical tolerance. Any instability in the inversion would manifest as oscillatory artefacts in the spatial profiles, and none are observed in Figures 2–19.

4. Consistency across the parametric studies. Finally, the parametric studies in Sections 6.1–6.3 produce smooth, monotonic, and physically interpretable trends across all parameter ranges explored—nonlocal parameters, pulse durations, and fractional orders. The ordered hierarchy of the fractional operators (NPTHMD-CC > NPTHMD-C > NPTHMD-AB), the monotonic dependence on pulse duration, and the systematic amplification by nonlocality are all consistent with physical expectation and would not emerge from an unstable numerical scheme.

We are grateful to the reviewer for raising this point, which allows us to confirm that the numerical methodology is both standard and independently verified. We believe the combination of the Honig–Hirdes algorithm, the quantitative benchmark agreement to within 10%, and the internal consistency checks fully addresses the concern, and we hope the reviewer will agree that no further revision is necessary.

Comment 7-How are results compared with limiting cases (classical, local, integer) and what is new?

Response to Reviewer Comment 7

We sincerely thank the reviewer for this question, which addresses the two most fundamental aspects of the paper: whether the model correctly reduces to established limiting cases, and what genuinely new contribution it makes. We are pleased to note that both aspects are already fully documented in the revised manuscript, and we therefore believe no further additions are required. We respectfully draw the reviewer's attention to the relevant evidence.

1. Reduction to the classical, local, and integer-order limits. The model is analytically and numerically shown to reduce to established limiting cases in two complementary ways:

Analytically, Table 1 enumerates nine special cases that the framework recovers under specific parameter choices. The classical local integer-order case (NTMD: $\alpha = 1$, $\xi = 0$, $\tau = 0$, $\gamma_m = \gamma_n = 0$) corresponds to Case 6 of Table 1. The reduction is performed on the dimensional governing equations prior to the Laplace transform: setting $\gamma_m = \gamma_n = 0$ decouples the carrier and moisture equations, which—together with the homogeneous initial conditions and zero-flux boundary conditions—admit only the trivial solutions $N \equiv M \equiv 0$. The 8×8 algebraic system of Eq. (64) then contracts to a 4×4 system in the amplitudes of e^* and T , recovering a Lord–Shulman-type formulation.

Numerically, the reduced classical limit is solved and benchmarked against the published hollow-cylinder results of Jojare and Gaikwad [59], as reported in Section 6.4.3 and Table 4. The maximum relative deviations are 5.99% for temperature and 9.80% for radial stress across the full radial domain, with exact agreement at the loaded inner surface where both formulations enforce the same compressive boundary condition. This confirms that the framework correctly recovers established generalized thermoelasticity in the appropriate limit.

In addition, the model is verified to reduce to classical Fourier conduction when $\tau_q \rightarrow 0$ and $K^* \rightarrow 0$, and to the integer-order derivative when $\alpha \rightarrow 1$. When all limits are taken simultaneously, the entire system collapses to classical coupled thermoelasticity.

2. What is new. The novelty of the work is not the physical coupling itself—which, as noted in the revised Introduction, has been established for solid cylinders under local elasticity by Singh et al. [46] and Kaur et al. [55]—but the systematic generalization of that framework in three specific directions, each of which is quantified in the revised manuscript:

Nonlocal elasticity (spatial parameter ξ and temporal parameter τ), which introduces size-dependent and long-range microstructural effects absent from [46] and [55]. Table 5 quantifies the resulting amplification of the peak radial displacement (approximately 100%), moisture concentration (42%), hoop stress (22%), and radial stress (16%). Crucially, the manuscript also demonstrates that spatial and temporal nonlocality applied separately produce different—and in some cases opposite—effects on the stress fields, showing that the two parameters are not interchangeable.

Atangana–Baleanu fractional memory with a non-singular Mittag–Leffler kernel, which replaces the integer-order or Caputo-type derivatives of prior studies with a memory operator that is simultaneously non-singular and non-local. Table 7 quantifies the resulting damping hierarchy (NPTHMD-CC > NFPTHMD-C > NFPTHMD-AB) across all six coupled fields, with reductions ranging from 4% (temperature) to 13% (radial displacement).

Hollow-cylinder geometry with distinct inner and outer boundary conditions, which introduces wave-reflection and interference phenomena absent from solid-cylinder formulations.

3. Position relative to the limiting cases. The relationship between the new contributions and the validated classical baseline is quantified explicitly in Table 7: the departures of the nonlocal fractional model from the classical limit (up to 13% for displacement and 8% for hoop stress) are an order of magnitude larger than the residual deviation from the independent benchmark (5.99% and 9.80%), confirming that the enhancements are physically significant rather than numerical artefacts. This establishes both that the model is correct in its limiting behaviour and that it predicts measurable new physics beyond that limit.

We are grateful to the reviewer for this question, which allows us to confirm that the reduction to limiting cases and the novelty statement are already documented in the revised Introduction, Table 1, Section 6.4.3, Table 4, and Tables 5–7. We hope the reviewer will agree that no further revision is necessary.

Comment 8-Update references to include: Effects of variable thermal conductivity and magnetic field on the photo-thermoelastic wave propagation in hydro-microelongated semiconductor, *Acta Mechanica* 236 (4), 2509–2527; Stochastic wave propagation in magneto-thermoelastic materials subjected to the change in electrical and thermal conductivity, *AIP Advances* 14 (1)

Response to Reviewer Comment 8

We thank the reviewer for this helpful suggestion and for bringing these two studies to our attention. Both references have been incorporated into the revised manuscript and are cited where relevant. They strengthen the qualitative validation of the model by extending the consistency assessment to magneto-thermoelastic and stochastic wave propagation settings. We are grateful for the reviewer's guidance, which has helped improve the contextual grounding of our work, and we hope the reviewer will agree that this aspect of the manuscript is now fully addressed.

Additional Questions:

Originality: Does the paper contain new and significant information adequate to justify publication?: YES

Relationship to Literature: To maintain the paper's integrity, we encourage citing relevant papers from reputable sources, including HFF. Is any significant work ignored? If your review contains requests for self-citations, please briefly explain your criteria for selecting references and how it is situated in relation to HFF. We have a policy in place that requires self-citation references to be removed from the decision letter during the review process.: YES

Methodology: Is the paper's argument built on an appropriate base of theory, concepts, or other ideas? Has the research or equivalent intellectual work on which the paper is based been well designed? Are the methods employed appropriate?: YES

Results: Are results presented clearly and analysed appropriately? Do the conclusions adequately tie together the other elements of the paper?: YES

Quality of Communication: Does the paper clearly express its case, measured against the technical language of the field and the expected knowledge of the journal's readership? Has attention been paid to the clarity of expression and readability, such as sentence structure, jargon use, acronyms, etc.: YES

We are sincerely grateful to the reviewer for the constructive and rigorous evaluation, which has substantially improved the clarity, accuracy, and credibility of the manuscript. We hope the revised version now meets the standards of the journal, and we thank the reviewer once again for their valuable time and expertise.